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1 IP ADDRESS

1.1 what is an IP address?

IP stands for Internet Protocol. It is a logical address assigned to every device connected to a network. It will help devices to communicate from one end device to another end device. It is used to communicate over a WAN(**Wide Area Network**) or LAN(**Local Area Network**). There are two types of IP address **IPV4** and **IPV6**. It is further divided in Private and Public IP address

1.2 IPV4

IPV4 (Internet Protocol Version 4) is based on 32 bit address. It has 4 octets. Each octet contains 8 bits. The format of IPV4 is **192.168.1.1** It supports 4.3 Billion Unique Address. Due to limited addresses subnetting is used.

1.3 IPV6

IPV6 (Internet Protocol Version 6) is based on 128 bit address. It has **8** groups. Each group is based on **16 bits**.

The format of IPV6 is **2001:0eb8:85a1:0000:8a2e:0370:7334:0380**. IPV6 provides much larger address space than IPV4.

1.4 Difference between IPV4 and IPv6

Feature	IPV4	IPV6
Address Length	32 bit	128 bit
Address Format	Decimal	Hexadecimal
Example Format	192.168.1.x	2001:0eb8:85a1:0000:0370:7334:0380
Security	Basic needs IPsec separately	Built in IPsec Support
Number of Groups/Octets	4 octets	8 groups
Broadcast	Supports BroadCast	Not Supported Broadcast. Uses Multicast

1.5 Practical Task to check IP Address

1.5.1 Objective

Identify IPV4 and IPV6 addresses of a Linux System

1.5.2 Tool Used

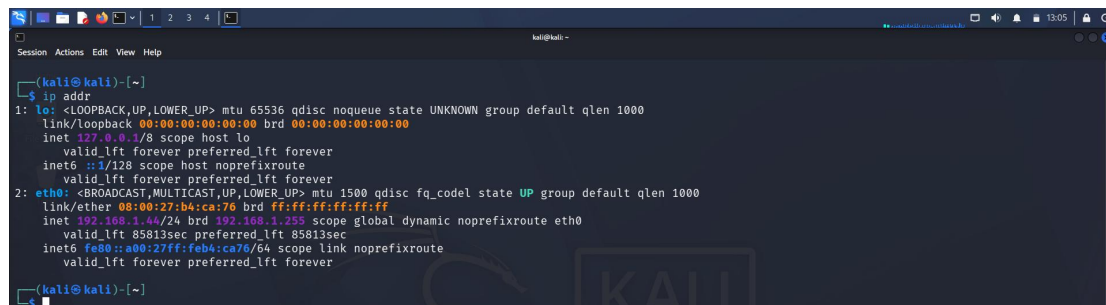
Kali Linux Terminal

1.5.3 Command Used

ip addr

1.5.4 Procedure

1. Open the Kali Linux terminal
2. Run the Command: **ip addr**
3. Find your network such as eth0 or wlan0
4. Find the IPV4 (**inet**) and IPV6 (**inet6**)



```
(kali@kali)~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:00:27:b4:ca:76 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.44/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 85813sec preferred_lft 85813sec
    inet6 fe80::a00:27ff:feb4:ca76/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

IPv4 **192.168.1.44/24** and IPv6 **fe80::a00:27ff:feb4:ca76/64** are displayed in above Picture .

2 MAC ADDRESS

2.1 what is a MAC Address?

MAC stands for **Media Access Control**. It is a unique Physical Address/ hardware address assigned to Machines NIC (**Network Interface Card**) by Manufacturer. It is permanent and Fixed Address Not changed like IP address. MAC address is used to communicate over a LAN (**Local Area Network**). It is also in hexadecimal format but in 6 groups. Example Format is **08:00:98:F4:CA:75**.

2.2 Purpose of a MAC Address

The main Purpose of a MAC address is to uniquely identify a device over a local area network for communication. With the help of MAC address message reached from one device to another device correctly.

2.3 Real World Example of a MAC Address

When a PC connects to a Wi-Fi router then router identifies it using its MAC Address before allowing communication over the LAN.

2.4 Practical Task to check MAC Address

2.4.1 Objective

Identify MAC Address of a Linux System

2.4.2 Tool Used

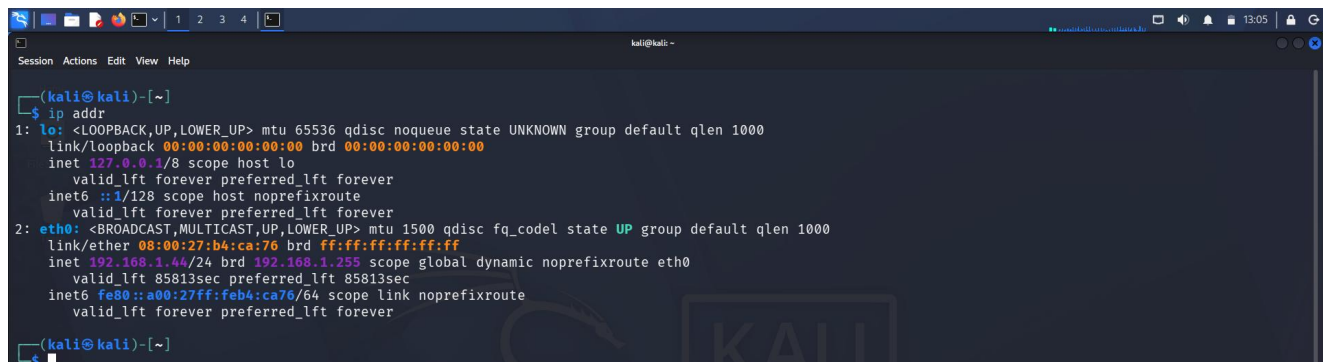
Kali Linux

2.4.3 Command Used

`ip addr`

2.4.4 Procedure

- 1) Open the kali Linux Terminal
- 2) Run the command: **ip addr**
- 3) Find your network interface such as `eth0` or `wlan0`.
- 4) Find the line starting with `link/ether` this is MAC address of a system



```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:b4:ca:76 brd ff:ff:ff:ff:ff:ff
   inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
       valid_lft 85813sec preferred_lft 85813sec
   inet6 fe80::a00:27ff:feb4:ca76/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
(kali@kali)-[~]
```

The MAC address of a system is shown in above picture which is **08:00:27:b4:ca:76**

3 OSI MODEL

3.1 what is the OSI Model?

OSI Stands for Open System Interconnection Model. It's a conceptual Model that explains how data travel from one device to another device with in a network. It contains seven layers each layer has its own responsibility.

3.2 Seven Layers of OSI Model

The seven Layers of OSI Model are listed below

Application Layer: Provides Network services directly to end users applications.

Example: Chrome, Gmail, WhatsApp, HTTP, HTTPS, DNS

Presentation Layer: This layer provides Compression, Encryption and Decryption of data for the application layer.

Example: SSL/TLS, JPG, PDF

Session Layer: This layer is used to establish, synchronize, and terminate the session between two applications.

Example: Session establishment, management and termination during a video call via Zoom, Google Meet etc.

Transport Layer: This layer provides end-to-end communication between different devices, flow control and error detection to ensure data moves in a reliable way. It supports two protocols UDP and TCP. Choosing between TCP and UDP completely depends on the situation what it is. Here Data is in the form of segments.

Example: Live video streaming of a cricket match is an example of transport layer that uses UDP protocol because here if some packets of data are lost then there is no issue we can tolerate this because in Live streams speed matters then perfection.

Network Layer: Receives a segment from Transport layer. Adds an IP Header then creates a Packet. Chooses the best path to route that Packet using a route table. Routers work here.

Example: Sending email over the Internet where router uses route table to efficiently route message to the correct destination.

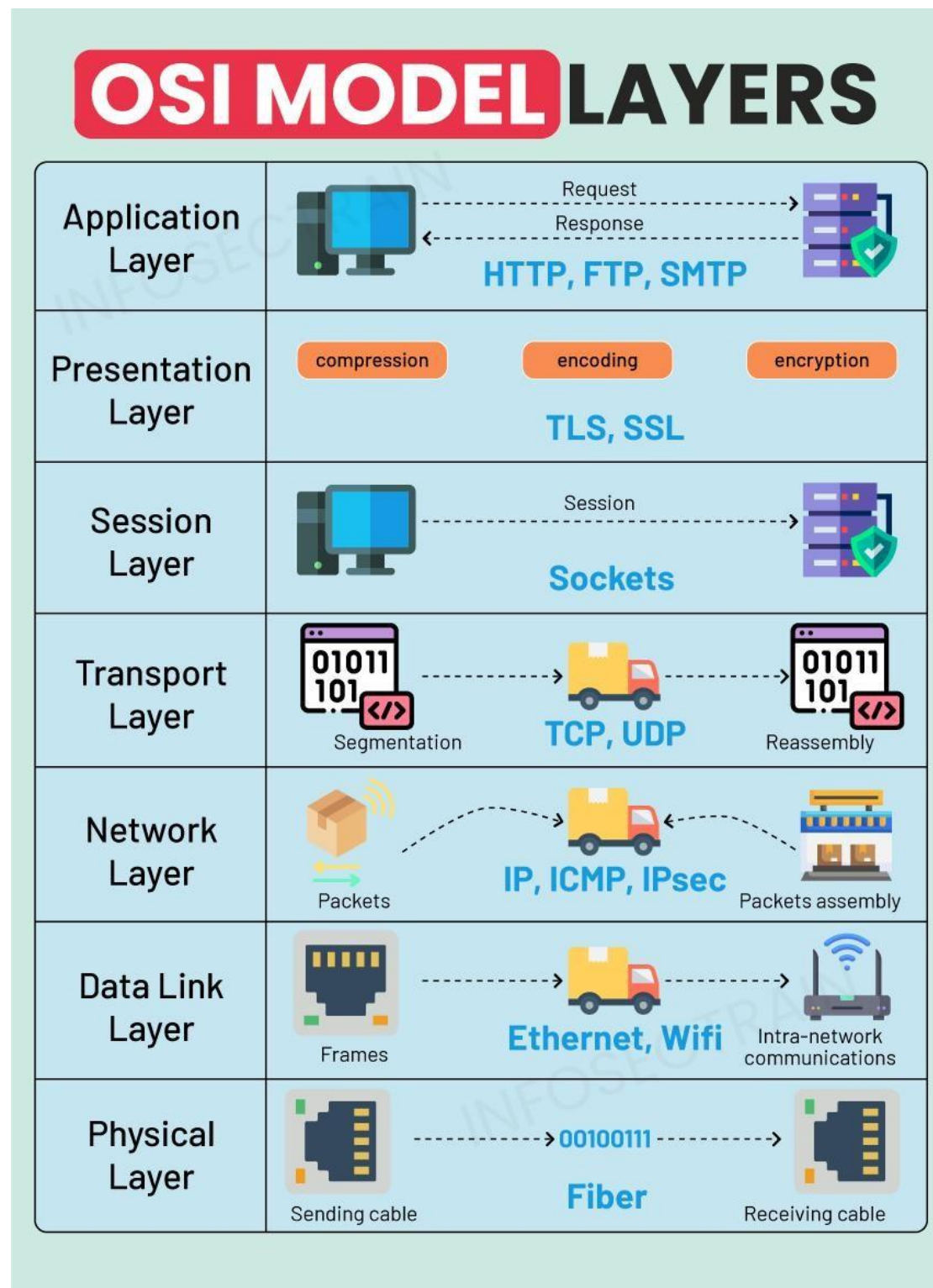
Data Link Layer: Data is received from network layer in Data Link Layer. Data Link Layer creates Frames. It adds a header to a Packet received from Network Layer. Source Mac and Destination Mac. Switches work here.

Example: Sharing a file from one laptop to another over a LAN the data reaches to correct destination after identifying the Source MAC and Destination MAC.

Physical Layer: Transmits raw bits over the Physical wire by encoding as electrical or optical signals. For wireless medium it uses radio waves.

Example: An Ethernet cable pulsing electricity or a Wi-Fi router emitting radio waves.

3.3 Pictorial Representation of OSI Model



4 Common Network Ports

4.1 What are Network Ports?

Network Ports are logical communication end point on which network application and services runs. There are total **0-65535** ports in computer Network. The IP address helps to identify the system but Port Number is required to direct data to the actual application or service running within that system. There are 3 major categories of ports

1. 0-1023:--> Well-known Ports/ System Ports
2. 1024-49151:--> Registered Ports/ User Ports
3. 49152-65535:--> Ephemeral Ports

4.2 Common Network Ports

S.No	Port Number	Port Functionality
1	21	Port use for FTP (File Transfer Protocol) service to share files between Client and Server
2	22	Port use for SSH(Secure Shell) service connection use for secure encrypted remote login on servers
3	25	Port use for SMTP (Simple Mail Transfer Protocol) service use to send emails
4	53	Port use for DNS (Domain Name System) service use to resolve Domain names like google.com into machine readable IP address .
5	80	Port use for HTTP (Hyper Text Transfer Protocol) use for web browsing Here traffic is not encrypted over the web
6	443	Port use for HTTPS (Hyper Text Transfer Protocol Secure) use for secure communication over the web communication is encrypted via TLS/SSL.
7	3389	Port use for RDP (Remote Desktop Protocol) service use for remotely accessing and controlling another computer's desktop over a network
8	23	Port use for Telnet Provides unencrypted remote access
9	110	Port use for POP3 (Post Office Protocol v3) use to receive incoming mails from a server
10	445	Port use for SMB (Server Message Block) service use to share files, printers over the network

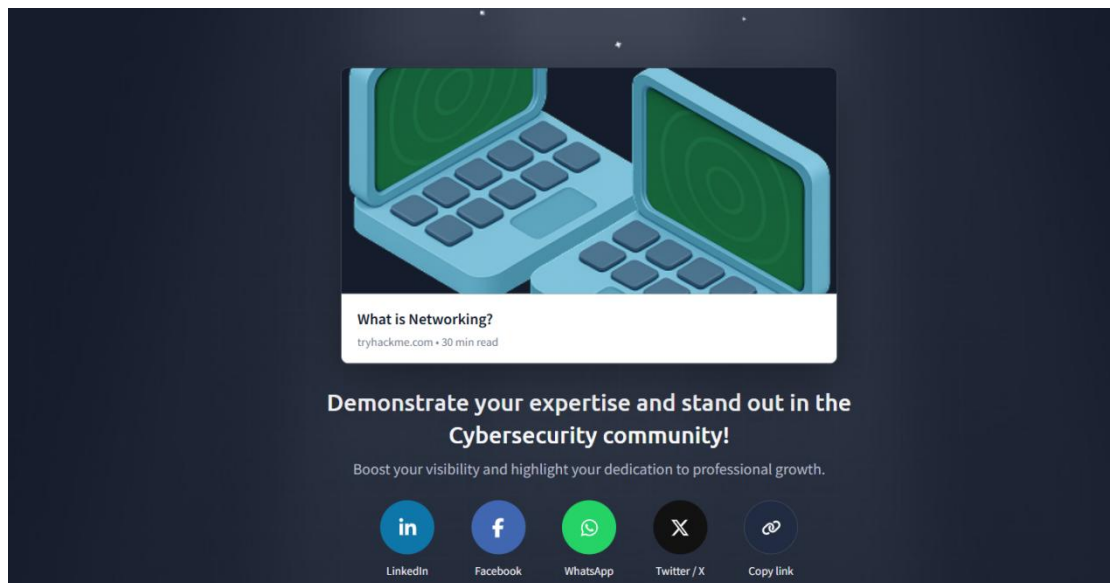
4.3 Practical Task

4.3.1 Objective

To understand how devices gets identified on the network with IP address how ping works to check system is alive or not and how spoofing MAC address will allow to visit a site that is blocked for specific MAC address

4.3.2 Platform Used

TryHackme **what is Networking room**



5 TCP VS UDP

5.1 What is TCP?

TCP stands for (**Transmission control protocol**) is a connection oriented protocol that ensures reliability and accuracy of data delivery. It checks for errors, packet delivery that's why it is slow as compare to UDP. It uses 3 way handshake

5.2 What is UDP?

UDP stands for (**User Datagram protocol**) is a connection less protocol that is simple and fast because it does not ensure reliability of data means some packets may be lost during transmission.

5.3 Difference between TCP and UDP

S.No	Functionality	TCP	UDP
1	Connection	Connection oriented	Connectionless
2	Error Checking	Yes	No
3	Speed	Slower than UDP	Faster than TCP
4	Reliability	Yes	No
5	Example	HTTP,HTTPS,SSH,SMTP	Live streaming, VoIP,DNS

6 Conclusion

This task strengthened my understanding of overall Networking Concepts as Networking is very crucial to understand before going deep into ethical hacking because attack also happens layer wise like on Data Link Layer , Network Layer etc. Attackers also uses different protocols of networking such as FTP,SSH,SMTP,SMB etc.

